

Big Data Research

The US State of Massachusetts announced the Big Data Initiative, to give state government and private funding for Big Data research.



Facebook's Project Prism

Facebook announced development of a new initiative - Prism - to help it better deal with the tonnes of data it generates everyday.

letting him learn more about the world he lives in. And this can happen in two ways – directly or indirectly.

Directly, Big Data can help you understand your surroundings better by providing new insights about everyday stuff that you might not be aware of normally. For example, consumers frequently pay their utility bills without realizing how much exactly each instrument in their home consumes. MacArthur Fellow Shwetak Patel decided to find a solution to this problem. He realized that most devices used today have unique digital signatures that can be detected with basic wireless sensors. Patel created an algorithm which, along with a sensor placed anywhere in the house, gives users inexpensive visual feedback about which devices use the greatest amount of electricity and how to conserve the wasted amount. Utilizing Big Data, Patel has empowered common consumers with the information they need to make better decisions that help them in improving the quality of their lives.



Big Data can help us learn stuff about ourselves we never imagined

Big Data analysis has also helped greatly in furthering knowledge in fields like the health sciences. Researchers John Guttag and Collin Stultz, along with Zee-shan Syed, have created a computer model to analyze discarded electro cardiography (EKG) data of heart attack patients, while Sheila Nirenberg of Weill Cornell Medical College has found a way to artificially recreate the complex code seen by a healthy retina upon seeing an image using massive amounts of data based on multiple studies and diagnoses of retinal diseases. The former stands to signifi-



Smarter Cities is an initiative by IBM which uses Big Data to assist city administrations

cantly improve today's risk-screening techniques for patients vulnerable to repeat heart attacks while the latter has led to the creation of custom software, embedded into microprocessors and cameras, that can be used to build eyeglasses that will help patients of severe retinal disease regain their vision.

Big Data can also be helpful in improving governance and creating better, smarter infrastructure. Information running into massive streams of data is generated daily by the digital utility services common in most cities today, which if properly harnessed can completely change the way authorities manage civic systems. Equipped with tools to analyze data for better decisions, anticipate problems to resolve them proactively and co-ordinate resources to operate effectively, smarter cities, and their administrations, are the drivers of economic growth and progress for society at large. The process of equipping cities with such facilities has already begun through initiatives like IBM's Smarter Cities. The IBM Smarter Cities initiative is an attempt to use Big Data to help make significant changes in the way city administrations approach their responsibilities. Utilizing the wealth of information revealed by Big Data about how cities tick, the initiative plans to make the jobs of officials easier and smoother, while giving citizens better standards of living through more efficient services. These are just some of the myriad

ways Big Data will help improve the world of everyone in general, not just the corporations and departments interested in data development. A conglomerate of individuals and companies recently launched The Human Face of Big Data initiative, in a bid to let common people understand how exactly Big Data can be useful to them. Using crowd-sourced information based on apps for smartphones, this initiative is helping people understand the wealth of information available to them through Big Data, and bringing them closer in the process. The apps contain a series of questions across categories, such as your Demographic, your opinions about friends, family, yourself, etc. Answering the questions truthfully creates a unique "signature" for you via your smartphone. The app then lets you compare your answers with the general trend of answers given by other users who've used the app on their smartphones, even finding your "data doppelganger" for you, someone whose

HOW BIG IS BIG DATA

- 247 billion emails are sent every day. (Up to 80 per cent are spam.)
- Every hour, we create enough Internet traffic to fill 7 billion DVDs. Side by side, that's seven times the height of Everest.
- There are 133 million blogs on the web.
- As of August 2012, there were just over 4 million articles in the English Wikipedia.